TEAM BOOGIE

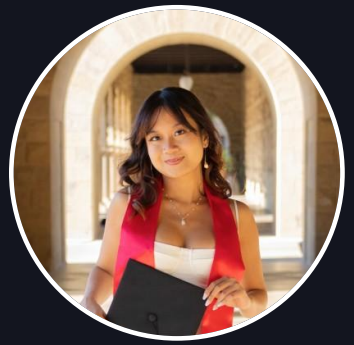
P6: User Studies

Ari B.
Osose E.
Mai H.
Rachel L.
Ananya N.

Meet the team

Ari B.

Osose E.



Mai H.

Rachel L.

Ananya N.

Study Method

Each interviewee completed two core tasks:

Task 1: **Schedule a Ride with BoogieBot**

→ communicate a pick-up location (inc. current location) & entrance

→ communicate a drop-off location & entrance

Task 2: **Explore Locations with Search**

→ search for a building by name

→ identify a desired entrance based on provided landmarks/keywords

Study Results (Quantitative)

	"Marcus"	"Adam"	"Mateo"
	Low vision Uses screen magnification	Blind Uses screen reader	Blind Uses screen reader
Task 1: Chatbot	4.50 min.	4.47 min.	11.52 min.
Task 2: Search	1.02 min.	1.45 min.	1.32 min.
Total	**5.52 min.**	**6.32 min.**	**13.24 min.**

Study Results (Qualitative → Critical Incidents)

"I'll be surprised if the search can find CoDa… oh wow!" 😀

Wheelchair button should be toggled "Off" by default ♿

"I don't think I would go straight to the chatbot for ride booking…" 💬

Chatbot seems geared towards exploration 🧭, while search is for the everyday 📍

"[Using cardinal directions] seems like the most 'correct' way to approach this problem" ✅

Major landmarks + directions + surrounding features together work well 👍

Conclusions

Takeaway #1

Overall, our prototype is a more usable and accessible tool than the existing DisGo/Flexigo application.

Search, landmarks, and chatbot run the gamut of abilities and skills

Takeaway #2

The range of interaction techniques needs to be accounted for: VoiceOver, dictation input, typing, Braille input, TalkBack, hybrid methods, etc.

The visual interface itself also still plays an important role in the usability.

Takeaway #3

The mental model of the user and the text-based model of the system need to align, so further refinement of the system's interpretation and parsing skills is still a prevalent ongoing task.

Next steps: Revisions and Reiteration

UI Color Contrast & Layout Dynamics

Low Vision Users
Examine the magnification effects on the app layout

Co-develop a color scheme that addresses varying sensitivity to light

Prompt Engineering

Chatbot Flexibility
Create a catch-all system to allow users to give information through a conversation or in a single message that is parsed

–OR–

Provide more explicit scaffolding for what the system expects of the user

Stronger LLM + Metadata Collaboration

Information Retrieval
Continue to increase the scope of the database

–AND–

Improve the chatbot's ability to logically and contextually process input (e.g. "Green Library coffee shop" = Coupa Cafe)

Perhaps query a map API to access additional context?

Default Settings & Error Prevention

Setting Fine-Tuning
Intentionally set default settings for ride-booking and other requirements (e.g. wheelchair accessibility)

Navigation
Test edge cases and unanticipated transitions between screens to ensure error-catching mechanisms are in place (e.g. disabled buttons)

Thank you!

Questions?